Word Order Variation in Spoken and Written Corpora

A Cross-Linguistic Study of SVO and Alternative Orders

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Introduction

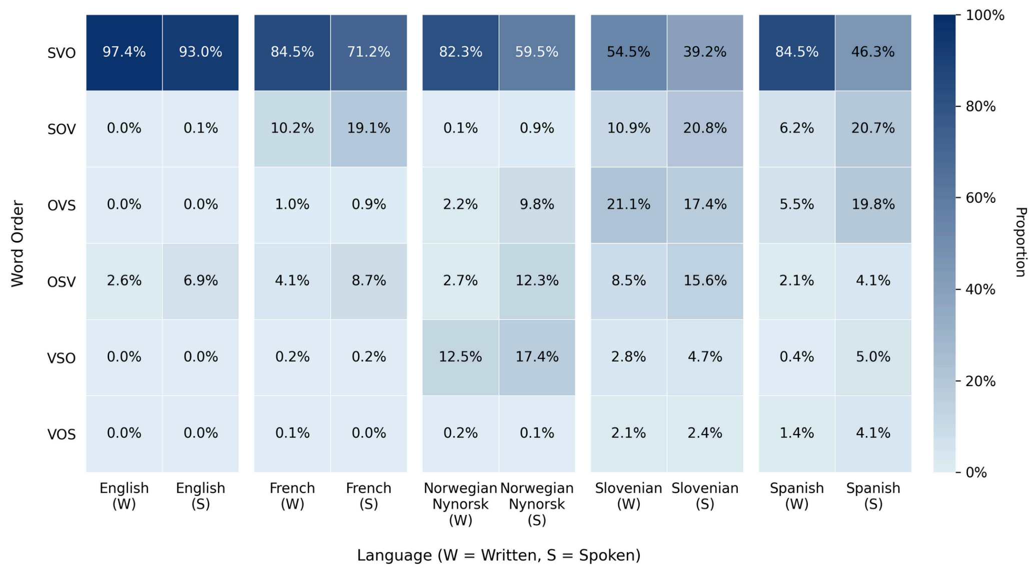
- Word order is central in linguistic typology (WALS, Grambank) and grammar research.
- Most typological generalisations come from written data, missing modality effects (e.g., writing vs. speech).
- Recent corpus studies show word order is **gradient** and **context-sensitive**.
- Gap: Spoken word order patterns are rarely studied cross-linguistically.

Objective

- Compare **spoken** vs. **written** SVO word order (Subject-Verb-Object) across **five Indo-European languages** (English, French, Norwegian Nynorsk, Slovenian, Spanish).
- Research questions:
  - **RQ1**: *Does word order differ between spoken and written language?*
  - **RQ2**: *If so, how does it differ?*
  - **RQ3**: *Are these differences similar across languages?*

Results and Analysis

Figure 1. Word Order Patterns by Language and Modality



General observations:

- **Spoken vs. written difference:** Word order patterns differ between speech and writing across all examined languages.
- **Greater variation in speech:** Spoken corpora show more diverse word orders, with SVO less dominant than in writing.
- **Cross-linguistic flexibility:** The degree of variation differs by language (greatest in Spanish and Slovenian, smallest in English).

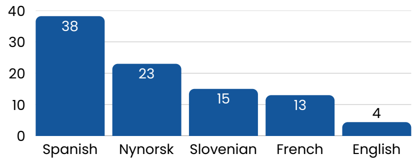


Figure 2. SVO Proportion Drop: Written to Spoken

Potential factors influencing the variation in speech:

- **Morphological richness:** Inflection reduces reliance on fixed word order.
- **Prosodic structuring:** Intonation, stress and rhythm support non-canonical sequencing.
- **Interaction-specific constructions:** Frequent use of pronouns, clitics, and questions give rise to alternative patterns.
- **Information structure:** Topic-focus ordering is more central in real-time production (e.g., object-fronting, postverbal subjects).

Methodology

- **Data:**
  - Spoken/written UD treebanks (v2.15), matched per language.
  - Corpora: **GUM** (split into spoken/written subsets), **Rhapsodie/GSD**, **NynorskLIA/Nynorsk**, **SST/SSJ**, **COSER/GSD**.
- **Data extraction**: Used STARK tool to retrieve all clauses with finite verb + nominal subject + direct object, regardless of clause type.
- **Classification**: Assigned each clause to one of six canonical word orders: SVO, SOV, VSO, VOS, OSV, OVS.
- **Analysis**: Compared order distributions across spoken and written modalities for each language.

Table 1. Canonical Word Order Examples in Slovenian. All sentences translate as 'Mother buys apples'.

Order	Slovenian	Gloss
SVO	*Mama kupuje jabolka*	mother buys apples
SOV	*Mama jabolka kupuje*	mother apples buys
VSO	*Kupuje mama jabolka*	buys mother apples
VOS	*Kupuje jabolka mama*	buys apples mother
OSV	*Jabolka mama kupuje*	apples mother buys
OVS	*Jabolka kupuje mama*	apples buys mother

Conclusion

- Modality strongly affects word order flexibility.
- Findings challenge written-based typologies and call for integrating spoken corpora.
- Future work: more languages, in-depth analysis, finer clause type distinctions, new metrics (e.g., entropy).

This work was financially supported by the Slovenian Research and Innovation Agency through the research project Treebank-Driven Approach to the Study of Spoken Slovenian (Z6-4617), and the research program Language Resources and Technologies for Slovene (P6-0411).